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Q1) Prove Fermat's Little theorem and use it to compute $a^{p-1} \bmod p$ for given values of $a=7$, $p=13$. Then, discuss how this theorem is useful in cryptographic algorithms like RSA.

Ans: Theorem:

If p is a prime number and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$.

Proof: Consider the integers,

$1, 2, 3, 4, \dots, (p-1)$, they leave the remainders

$1, 2, 3, \dots, (p-1)$ when divided by p .

Consider an integer a , relatively prime to p .

$a \cdot 1, a \cdot 2, a \cdot 3, \dots, a \cdot (p-1)$

Observe that if,

$$a \cdot i \equiv a \cdot j \pmod{p}, \quad 1 \leq i, j \leq p-1$$

then $a(i-j) \equiv 0 \pmod{p}$

which is true if and only if $i=j$

$\therefore a_i \not\equiv a_j \pmod{p}$, for $1 \leq i, j \leq p-1, i \neq j$

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$\therefore a \cdot 1, a \cdot 2, a \cdot 3, \dots, a \cdot (p-1)$ leaves $p-1$ number of different remainders when divided by p , that is $a \cdot 1, a \cdot 2, a \cdot 3, \dots, a \cdot (p-1)$ are congruent to $1, 2, 3, \dots, p-1$ but in some order.

$$a \cdot 1 \cdot a \cdot 2 \cdot a \cdot 3 \cdot \dots \cdot a \cdot (p-1) \equiv 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-1) \pmod{p}$$

$$a^{p-1} (p-1)! \equiv (p-1)! \pmod{p}$$

$$\Rightarrow (p-1)! \{a^{p-1} - 1\} \equiv 0 \pmod{p}$$

$$\Rightarrow a^{p-1} - 1 \equiv 0 \pmod{p}$$

$$\Rightarrow a^{p-1} \equiv 1 \pmod{p} \quad (1, 2, 3, \dots, p-1)$$

Given that,

$$a = 7$$

$$p = 13$$

We know that, $a^{p-1} \equiv 1 \pmod{p}$

$$\Rightarrow 7^{13-1} \equiv 1 \pmod{13}$$

$$\Rightarrow 7^{12} \pmod{13} = 1$$

$$\Rightarrow 7^{12} \pmod{13} = 1$$

Fermat's Little Theorem plays a vital role in modern cryptography, especially in RSA, by enabling secure and efficient modular exponentiation.

Use in RSA algorithm:

RSA encryption and decryption involve raising numbers to large powers modulo n , where $n = p \times q$.

Fermat's Theorem helps in two key ways:

1. Efficiency in Computation:

Instead of directly computing $a^k \bmod p$, Fermat's theorem simplifies calculations using:

$$a^{p-1} \equiv 1 \bmod p$$

This helps reduce the exponent modulo $p-1$, making operations faster.

2. Foundation of Reversibility:

RSA ensures that: $(M^e)^d \equiv M \bmod n$

Fermat's theorem guarantees that the decryption operation returns the original message correctly by using

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properties of mod arithmetic.

Fermat's Theorem contributes to the mathematical backbone of RSA, ensuring:

- correctness of encryption/decryption,
- Efficient computation,
- Resistance to brute force attacks due to large number handling.

Q2) Euler Totient Function: Compute $\phi(n)$ for $n = 35, 45, 100$. prove that if a and n are co-prime, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Ans: $\phi(35) = \phi(7 \times 5)$

$$= \phi(7) \times \phi(5)$$

$$= 6 \times 4$$

$$= 24$$

$$\phi(45) = \phi(3^2 \times 5)$$

$$= \phi(3^2) \times \phi(5)$$

$$= (3^2 - 3^{2-1}) \times 4$$

$$= (9 - 3) \times 4 = 24$$

$$\begin{array}{r} 3 \overline{) 45} \\ 3 \overline{) 15} \\ 5 \end{array}$$

$$= 3^2 \times 5$$

$$\phi(100) = \phi(2^2 \times 5^2)$$

$$= \phi(2^2) \times \phi(5^2)$$

$$= (2^2 - 2^1) \times (5^2 - 5^1)$$

$$= (4 - 2) \times (25 - 5)$$

$$= 2 \times 20$$

$$= 40$$

$$\begin{array}{r} 2 \overline{) 100} \\ \underline{200} \\ 50 \\ 2 \overline{) 50} \\ \underline{20} \\ 30 \\ 5 \overline{) 30} \\ \underline{25} \\ 5 \\ 5 \overline{) 5} \\ \underline{5} \\ 0 \end{array}$$

Statement: If n is a positive integer and a

be any integer relatively prime to n , then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where ϕ is the Euler ϕ function.

Proof: Let $[x]$ denote the residue class of the set of integers mod n . Let $G = [a] : a \text{ is an integer relatively prime to } n$.

Then we know that w.r.t multiplication of residue classes G is a group of order $\phi(n)$. The identity element of this group be the residue class $[1]$.

$$\text{we have, } [a] \in G \Rightarrow [a]^{\phi(n)} = [1]$$

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$$\Rightarrow [a], [a], [a], \dots \text{ up to } n \text{ times} = [1]$$

$$\Rightarrow [aa, \dots \text{ up to } \phi(n) \text{ times}] = [1]$$

$$\Rightarrow [a^{\phi(n)}] = [1]$$

$$\Rightarrow a^{\phi(n)} = 1 \pmod{n}$$

Q3) Solve the system of congruences using the Chinese Remainder Theorem and prove that x congruent to 11 on mod $N = 3 \times 4 \times 5 = 60$.

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{4}, x \equiv 1 \pmod{5}$$

Ans:

The Chinese Remainder Theorem (CRT) is used to solve a set of different congruent equations with one variable but different moduli which are relatively prime as shown below:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_n \pmod{m_n}$$

CRT states that the above equations have a unique solution if the moduli are relatively prime.

$$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_n M_n M_n^{-1}) \pmod{M}$$

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$$x \equiv 2 \pmod{3}$$

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv 1 \pmod{5}$$

$$x \equiv a_3 \pmod{m_3}$$

$$a_1 = 2, a_2 = 3, a_3 = 1$$

$$m_1 = 3, m_2 = 4, m_3 = 5$$

$$M = m_1 \times m_2 \times m_3$$

$$= 3 \times 4 \times 5$$

$$= 60$$

$$M_1 = \frac{M}{m_1}$$

$$= \frac{60}{3}$$

$$= 20$$

$$M_2 = \frac{M}{m_2}$$

$$= \frac{60}{4}$$

$$= 15$$

$$M_3 = \frac{M}{m_3}$$

$$= \frac{60}{5}$$

$$= 12$$

$$M_1 \times M_1^{-1} = 1 \pmod{m_1}$$

$$20 \times 2 = 1 \pmod{3}$$

$$M_1^{-1} = 2$$

$$M_2 \times M_2^{-1} = 1 \pmod{m_2}$$

$$15 \times 3 = 1 \pmod{4}$$

$$M_2^{-1} = 3$$

$$M_3 \times M_3^{-1} = 1 \pmod{m_3}$$

$$12 \times 8 = 1 \pmod{5}$$

$$12 \times 8 = 1 \pmod{5}$$

$$M_3^{-1} = 8$$

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$$x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}) \bmod M$$

$$= (2 \times 20 \times 2 + 3 \times 15 \times 3 + 12 \times 12 \times 8) \bmod 60$$

$$= (80 + 135 + 144) \bmod 60$$

$$= 311 \bmod 60$$

$$= 47$$

$$\text{So, } x \equiv 11 \pmod{60}$$

$x = 11$ satisfies all three congruences.

Q4) Find whether 561 is a Carmichael number by checking its divisibility and Fermat's test.

Ans: The composite number n is a Carmichael number if, whenever a is relatively prime to n , we have,

$$a^{n-1} \equiv 1 \pmod{n}$$

561 is a Carmichael number.

1) The prime factorization of 561 is

$$561 = 3 \times 11 \times 17$$

So, 561 is composite.

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$$1) a^{560} \equiv 1 \pmod{561}, \text{ if } (a, 561) = 1$$

$$\text{we have, } 561 = 3 \times 11 \times 17$$

$$(a, 561) = 1 \Rightarrow 3 \nmid a$$

$$\text{Similarly, } 11 \nmid a \rightarrow (11, a) = 1 \text{ and}$$

$$17 \nmid a \rightarrow (17, a) = 1$$

Now by Fermat's theorem,

$$3 \text{ is a prime with } (3, a) = 1$$

$$\rightarrow a^2 \equiv 1 \pmod{3}$$

$$(a^2)^{280} \equiv 1 \pmod{3}$$

$$a^{560} \equiv 1 \pmod{3} \text{ --- (i)}$$

$$\text{Similarly, } 11 \text{ is a prime with } (11, a) = 1$$

$$\rightarrow a^{10} \equiv 1 \pmod{11}$$

$$\rightarrow (a^{10})^{56} \equiv 1 \pmod{11}$$

$$a^{560} \equiv 1 \pmod{11} \text{ --- (ii)}$$

$$17 \text{ is a prime with } (17, a) = 1$$

$$\rightarrow a^{16} \equiv 1 \pmod{17}$$

$$(a^{16})^{35} \equiv 1 \pmod{17}$$

$$a^{560} \equiv 1 \pmod{17} \text{ --- (iii)}$$

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Since 3, 11 and 17 are distinct prime and are relatively prime to another,

from (i), (ii) and (iii)

$$a^{560} \equiv 1 \pmod{3 \cdot 11 \cdot 17}$$

$$a^{560} \equiv 1 \pmod{561}$$

Thus by definition of Carmichael number,

561 is a Carmichael number.

Q5) Find a generator (primitive root) of the multiplicative group modulo 17.

Primitive root: A number a is a primitive root modulo n if every number coprime to n is congruent to a power of a modulo n .

In a simple sentence,

a is said to be a primitive root of prime number p , if $a \pmod{p}$, $a^2 \pmod{p}$, $a^3 \pmod{p}$, $a^4 \pmod{p}$ are distinct.

2 is not a primitive root of modulo 17.

Because,

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$$2^1 = 2 \bmod 17$$
$$= 2$$

$$2^9 = 512 \bmod 17$$
$$= 2$$

$$3^1 = 3 \bmod 17$$
$$= 3$$

$$3^2 = 9 \bmod 17$$
$$= 9$$

$$3^3 = 27 \bmod 17$$
$$= 10$$

$$3^4 = 81 \bmod 17$$
$$= 13$$

$$3^5 = 243 \bmod 17$$
$$= 5$$

$$3^6 = 729 \bmod 17$$
$$= 15$$

$$3^7 = 2187 \bmod 17$$
$$= 11$$

$$3^8 = 6561 \bmod 17$$
$$= 16$$

Here, 2 is not distinct value, so
2 is not a primitive root.

$$3^9 = 19683 \bmod 17$$
$$= 14$$

$$3^{10} = 59049 \bmod 17$$
$$= 8$$

$$3^{11} = 177147 \bmod 17$$
$$= 7$$

$$3^{12} = 531441 \bmod 17$$
$$= 4$$

$$3^{13} = 1594323 \bmod 17$$
$$= 12$$

$$3^{14} = 4782969 \bmod 17$$
$$= 2$$

$$3^{15} = 14348907 \bmod 17$$
$$= 6$$

$$3^{16} = 43046721 \bmod 17$$
$$= 1$$

So, 3 is a primitive
root of modulo 17.

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Q6) Solve the Discrete Logarithm problem

Find x such that $3^x \equiv 13 \pmod{17}$

Ans: We can do this by computing the power of 3 modulo 17 until we reach 13.

$$3^1 \equiv 3 \pmod{17} = 3$$

$$3^2 \equiv 9 \pmod{17} = 9$$

$$3^3 \equiv 27 \pmod{17} = 10$$

$$3^4 \equiv 81 \pmod{17} = 13$$

From the calculations, we can see that,

$$3^4 \equiv 13 \pmod{17}.$$

Therefore $x = 4$.

Q7) Discuss the role of the discrete logarithm in the Diffie-Hellman key Exchange.

Ans: Role of discrete logarithm in Diffie-Hellman key Exchange -

1. Public parameters; Large prime P , generator g .

2. Key Exchange:

- Alice sends $A = g^a \bmod P$

- Bob sends $B = g^b \bmod P$

- shared key: $g = g^{ab} \bmod P$

3. Discrete logarithm Problem (DLP):

• Hard to find a from $A = g^a \bmod P$

• This difficulty ensures security.

4. Attacker's challenge:

• cannot compute shared key without solving DLP.

• DLP is computationally hard for large P .

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Q8) Compare and contrast the substitution cipher, Transposition cipher and playfair cipher.

Ans:

1. Substitution cipher:

Encryption Mechanism: Each letter is replaced by another letter.

Example: Caesar cipher shifts each letter by a fixed number.

Key space: For monoalphabetic: $26! \approx 4 \times 10^2$

2. Transposition cipher:

Encryption Mechanism:

- Letters are rearranged based on a pattern or key.
- No change to actual letters.

3. Playfair cipher:

Encryption Mechanism:

- Encrypt digraphs (pairs of letters)
- Use rules: same row, column or rectangle.

$$Q9) E(x) = (ax + b) \bmod 26, a=5, b=8$$

(a) Encrypt the plaintext "Dept of ICT, MBSTU"

Step A: Encryption

1. Preprocessing the plaintext:

Remove punctuation and spaces, convert to uppercase:

plaintext = "DEPTOFICTMBSTU"

2. Convert letters to numbers

D=3, E=4, P=15, T=19, O=14, F=5, I=8, C=2, T=19,

M=12, B=1, S=18, T=19, U=20

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3. Letter X E(X) Cipher

D 3 $(5 \times 3 + 8) \% 26$ X
= 23

E 4 $(5 \times 4 + 8) \% 26$ C
= 2

P 15 $(5 \times 15 + 8) \% 26$ B

T 19 $(5 \times 19 + 8) \% 26$ ✓
= 21

O 14 $(5 \times 14 + 8) \% 26$ A
= 0

F 5 $(5 \times 5 + 8) \% 26$ H
= 7

W 8 $(5 \times 8 + 8) \% 26$ W
= 22

C 2 $(5 \times 2 + 8) \% 26$ S
= 18

T 19 $(5 \times 19 + 8) \% 26$ V
= 21

M 12 $(5 \times 12 + 8) \% 26$ Q
= 16

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$$B \quad 1 \quad (5 \times 1 + 8) \% 26 \\ = 13$$

$$S \quad 18 \quad (5 \times 18 + 8) \% 26 \\ = 16$$

$$T \quad 19 \quad (5 \times 19 + 8) \% 26 \\ = 21$$

$$U \quad 20 \quad (5 \times 20 + 8) \% 26 \\ = 6$$

Step b: Decryption

The decryption function of Affine cipher is:

$$D(y) = a^{-1} \cdot (y - b) \bmod(26)$$

where a^{-1} is the modular inverse of $a = 5$ modulo 26.

$$\text{Since: } 5 \cdot 21 \equiv 105 \equiv 1 \bmod 26 \Rightarrow a^{-1} = 21$$

So, the decryption function becomes:

$$D(y) = 21 \cdot (y - 8) \bmod(26)$$

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2. Apply decryption on ciphertext:

Ciphertext: XCBVAHWSVQNGVG

Convert letters to numbers:

X = 23, C = 2, B = 1, V = 21, A = 0, H = 7, W = 22, S = 18, V = 21, G = 6

Q = 16, N = 13, Q = 16, V = 21, G = 6

Apply $D(y) = 21 \cdot (y - 8) \bmod 26$:

Letter	$\frac{y}{3}$	$D(y)$	Plain
X	23	$21 \times (23 - 8) \bmod 26 = 3$	D
C	2	$21 \times (2 - 8) \bmod 26 = 4$	E
B	1	$21 \times (1 - 8) \bmod 26 = 15$	P
V	21	$21 \times (21 - 8) \bmod 26 = 19$	T
A	0	$21 \times (0 - 8) \bmod 26 = 14$	O
H	7	$21 \times (7 - 8) \bmod 26 = 5$	F
W	22	$21 \times (22 - 8) \bmod 26 = 8$	I
S	18	$21 \times (18 - 8) \bmod 26 = 2$	C
V	21	$21 \times (21 - 8) \bmod 26 = 19$	T
Q	16	$21 \times (16 - 8) \bmod 26 = 12$	M
N	13	$21 \times (13 - 8) \bmod 26 = 11$	B
Q	16	$21 \times (16 - 8) \bmod 26 = 12$	S
V	21	$21 \times (21 - 8) \bmod 26 = 19$	T
G	6	$21 \times (6 - 8) \bmod 26 = 20$	V

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Cipher text: XCBUAHWSVQNGUG

Decrypted text: DEPTOFICTMBSTU

Q10) Design a simple novel cipher.

Ans:

Substitution: Each character is substitution using a keyed caesar shift.

Permutation: Blocks of text are permuted using a PRNG-based shuffle.

PRNG: Custom Linear congruential generator

Key:

k_1 : Integer

k_2 : seed value for PRNG

Block size: Fixed Block size

Encryption process:

Step 1: Substitution:-

Each character c in plaintext is shifted forward using a caesar-like method with a varying

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shift based on the PRNG.

PRNG: $X_{n+1} = (a \cdot X_n + c) \bmod m$

Example: Inputs:

Plaintext: "Hello"

$k_1 = 3; k_2 = 7$, Block Size = 2

Step 1: Substitution

Let's say PRNG gives shift = [5, 12, 7, 19, 2]

H $\rightarrow$ H(7) + 5 + 3 = 15 $\rightarrow$ P

E $\rightarrow$ E(4) + 12 + 3 = 19 $\rightarrow$ T

L $\rightarrow$ L(11) + 7 + 3 = 21 $\rightarrow$ V

L $\rightarrow$ L(11) + 19 + 3 = 33 $\rightarrow$ H (mod 26)

O $\rightarrow$ O(14) + 2 + 3 = 19 $\rightarrow$ T

Substituted: "PTVHT"

Step 2: Permutation (Block Size 2)

split: [PT] [VH] [T_]

Final ciphertext: "TPHV-T"